

Thermal Fractional Ore Processing System

Space Resource Extraction Application — v5

Executive Summary

The space variant of the Thermal Fractional Ore Processing System applies the same architectural principles — staged thermal separation, closed-loop resource recovery, spectroscopic process control, and zero-waste output — to asteroid, lunar, and planetary body resource extraction. The v5 design significantly improves space transferability: EM (Lorentz force) separation is gravity-independent by physics, hydrogen reduction chemistry works in any gravitational field, and sub-atmospheric Bay 3 operation translates naturally to pressure-vessel operation in vacuum environments.

Environments Addressed

- Low gravity bodies — Moon (0.16g), Mars (0.38g), large asteroids
- Zero gravity — orbital facilities, small asteroids, microgravity processing
- Vacuum — asteroid surface, lunar surface, orbital
- Controlled atmosphere — sealed processing with tunable gas composition
- Combinations — addressed by module substitution, not redesign

What Remains Constant Across All Environments

- Thermal cascade — heat stages material through melt thresholds regardless of gravity

- LIBS spectroscopic monitoring — electromagnetic, unaffected by gravity or vacuum
- EM (Lorentz force) separation — $F = J \times B$ has no gravity term. Operates identically from Earth surface to orbit.
- Hydrogen reduction — $\text{FeO} + \text{H}_2 \rightarrow \text{Fe} + \text{H}_2\text{O}$ works in any gravitational field. Product is water, not CO_2 .
- Off-gas capture — pressure differential moves gas regardless of gravity
- Sub-atmospheric Bay 3 — 0.5 atm has natural synergy with vacuum environments
- Modular bay architecture — same interface, different modules per environment
- Provenance stamping — works on any solidified output in any environment
- Zero waste philosophy — same product hierarchy, catch basin, second-pass architecture

EM Separation as Primary Architecture

The EM separator applies electromagnetic force to conductive liquid metals while non-conductive silicate slag is unaffected. This is inherently gravity-independent — the force acts on conductivity, not mass or density. The separator has no moving parts, no bearings, no seals to fail in vacuum, and no mechanical contact with the melt.

- No centrifuge bearings to degrade in vacuum or under thermal cycling
- No blast containment required (critical mass/volume savings in space)
- No vibration transmission to facility structure
- Operates at any gravity from 0g to 1g without modification
- Cu recovery: 98–99% regardless of gravitational environment

In the original design, the centrifuge was the most complex gravity-sensitive component. Its replacement with EM separation eliminates the single largest adaptation barrier for

space deployment.

Hydrogen Reduction in Space

The terrestrial system uses hydrogen as both heating fuel and chemical reductant at Bay 3. This chemistry — $\text{FeO} + \text{H}_2 \rightarrow \text{Fe} + \text{H}_2\text{O}$ — is directly transferable to space. Hydrogen is producible from water via electrolysis using solar power. The reaction product is water, which can be recycled back to the electrolyzer for more hydrogen. This creates a closed hydrogen/water loop that is self-sustaining given solar energy and an initial water supply.

Carbonaceous chondrite asteroids and lunar polar ice provide water for this loop. The ore body supplies its own processing consumable, making the system operationally self-sustaining at interplanetary distances.

Sub-Atmospheric Operation and Vacuum Synergy

Bay 3 operates at 0.5 atm in the terrestrial design. In space, the external pressure is effectively zero — Bay 3 operates at a modest positive pressure relative to vacuum surroundings, simplifying sealing and gas containment versus full atmospheric pressure.

The vacuum environment also enables vacuum distillation as a parallel separation mechanism. Materials evaporate at temperatures far below their atmospheric boiling points in vacuum. This creates a separation cascade alongside the thermal melt cascade — materials that resist melt separation may separate readily by vapor pressure.

Vacuum Distillation

- Zinc — boils at 907°C at 1 atm, evaporates at much lower temperatures in vacuum

- Lead, cadmium, mercury, arsenic, antimony — usable vapor pressures at achievable temperatures
- Sulfur compounds — volatile in vacuum at temperatures where they are solid in atmosphere

Space provides cold condenser surfaces for free. Panels oriented toward deep space reach temperatures far below any Earth-surface cooling system. The thermal gradient for condensation is orders of magnitude larger than on Earth.

Gravitational Environment Analysis

Full Gravity (Earth — 1g)

Baseline terrestrial system. Gravity drainage for melt flow. EM separation at Bay 3. Passive density stratification at other bays. Centrifuge modules where needed for non-conductive mineral pairs.

Reduced Gravity (Moon 0.16g, Mars 0.38g)

Gravity drainage slower. Wider drain apertures, longer residence times. EM separation unaffected — gravity-independent by physics. Centrifuge modules preferred at most bays for density separation of non-conductive fractions.

Zero Gravity (Orbital, Small Asteroids)

No gravity drainage. EM separation fully functional. Melt flow uses pressure differential (sealed bays with gas pressure) or electromagnetic pumping for conductive melts. Centrifuge bays for non-conductive density separation. Surface tension is dominant containment force.

Melt Flow Without Gravity

- Pressure differential flow: sealed bays with gas pressure from one side. Off-gas infrastructure provides pressurized gas.

- Electromagnetic pumping: conductive melts respond to moving magnetic fields. Natural extension of EM separator technology.
- Centrifugal bay design: bay spins to drive melt flow for non-conductive fractions.

Thermal Management in Space

Heat Retention

In vacuum, no convective heat loss — only radiative. Bay temperatures easier to maintain. Partially offsets energy cost of active flow systems.

Cooling and Quench

- Gas quench — inert gas injected, captured, recirculated
- Radiative shock — hot material exposed to space-cooled surface
- Mechanical contact — press against cold refractory surface
- Spray quench — water from asteroid hydrated minerals, flash-vaporizes, steam drives turbine before condensing

In-Situ Resource Utilization

Carbonaceous chondrite asteroids contain water ice, hydrated minerals, and carbon. Water provides hydrogen (via electrolysis) for both Bay 3 reduction and heating. The ore body supplies its own processing consumables.

Hydrogen/water closed loop: electrolyzer splits water → H₂ for Bay 3 reduction and combustion heating → product is H₂O → recycled to electrolyzer. Working fluid circulates indefinitely with minimal losses given solar energy input.

Atmospheric Options

- Inert — argon or nitrogen blanket, prevents oxidation

- Reducing — H₂ for metal oxide reduction (consistent with terrestrial Bay 3)
- Oxidizing — O₂ (from electrolysis) burns off sulfide minerals
- Vacuum — full vacuum distillation, lowest contamination

Individual bays operate under different atmospheres simultaneously. H₂ combustion heating in some bays, vacuum in others, inert blanket in others.

Configuration Matrix

- Earth surface (1g, Controlled): EM separation + gravity drain, H₂ combustion
- Moon surface (0.16g, Vacuum): EM separation + centrifuge, vacuum distillation, H₂ reduction
- Mars surface (0.38g, CO₂ atm): EM separation + partial gravity, H₂ reduction, atmospheric tuning
- Asteroid (~0g, Vacuum): EM separation + EM pumping, vacuum distillation, H₂ reduction
- Orbital (0g, Controlled): EM separation + EM pumping + pressure flow, full vacuum distillation

Output Handling in Space

Metal ingots, silicate fractions, and residue blocks produced identically to terrestrial. Provenance stamping unchanged. In zero gravity, units secured to storage structures. Uniform geometry simplifies automated handling.

Refractory concentrate — chromite, zirconia, alumina concentrated by two-pass system — has particular value for thermal protection systems, rocket nozzles, and structural components. Producing these from asteroid feedstock eliminates Earth launch costs.

Strategic Significance

The v5 design changes transform TFOPS from a system

requiring significant space adaptation to one where most hardware transfers directly:

- EM separation: eliminates the most complex gravity-sensitive component
- H₂ reduction: provides space-compatible reduction chemistry with closed water loop
- H₂ combustion heating: producible from water + solar in any environment
- 0.5 atm Bay 3: natural synergy with vacuum environments

The Architectural Insight

Most systems designed for one environment require fundamental redesign for another. TFOPS requires module substitution. The terrestrial system is not a precursor to the space system — it is the same system with environment-appropriate implementation modules installed. This is the direct consequence of building around thermodynamic and electromagnetic principles that do not depend on gravity, and around separation mechanisms (EM force, not centrifugal force) and reduction chemistry (H₂, not carbon) that are inherently gravity-independent.

A company deploying terrestrial TFOPS installations builds direct operational experience with every subsystem that the space variant requires. The technology development pathway is sequential and self-funding rather than requiring speculative capital for a system with no terrestrial revenue.